

Alfred Nonis

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EXPERIENCE

Longhorn Baja SAE — *Founding Member, Chassis Co-Lead*

June 2025 - PRESENT

- Designed a parametric off-road roll cage in SolidWorks around subsystem needs and three tube sizes, reducing chassis weight by 31%
- Validated chassis strength with ANSYS FEA under a 10,000 lbf frontal impact, 1,400 lbf shock load and engine-mount loading
- Redesigned chassis and subsystem mounts for easier fabrication, lower cost and available hardware
- Fabricated, jigged and tacked the chassis while integrating steering, powertrain, ergonomics, and suspension components

Texas Spacecraft Laboratory – *REACT Mission, Internal Hardware Lead*

April 2025 - PRESENT

- Leading a 3-person hardware team designing CubeSat electronics mounts, sensor hardware and prototype test equipment in SolidWorks
- Designed test-board hardware layout to control flywheel exposure to magnetic fields and support reliable sensor data collection
- Designed a simple liquid damping system to reduce vibration during prototype testing

Encotech Engineering Consultants, Inc. – *MEP & Social Media Intern*

June 2022 - August 2022

- Drafted MEP plans for an apartment amenity center, including HVAC layouts, and reviewed designs with senior engineers
- Managed social media accounts and researched competitors

EDUCATION

University of Texas at Austin – *B.S., Mechanical Engineering* Specializing in Design & Manufacturing and Energy Systems

August 2023 - August 2027

Courses Completed: Heat Transfer, Machine Elements, Mechatronics, Materials, Materials Processing, Fluid Dynamics, Dynamics, Matlab, Statistics

PROJECTS

Internal Combustion Engine Rebuild

Designed & 3D-printed a prototype intake velocity stack for a 4-stroke engine, recorded an average torque increase of 7.2% across all RPMs. Assembled a 2-stroke engine.

Go-Kart

Using Solidworks to develop a chassis by combining two shopping carts together to make a go kart. Currently trying to create a powertrain system while under heavy constraints.

SKILLS

Engineering Software

SolidWorks, Ansys,
Matlab, AutoDesk
Inventor, Revit,
ideaMaker, Bambu Studio

Hands-on

Worked on brake calipers,
pads, rotors, spark plugs,
oil/air filters, tire
rotation/replacement,
EVAP; woodworking,
laser cutting, TIG
welding, manual hole
saw, drop saw, band saw

Office Software

MS Word, MS Excel, MS
Powerpoint

Involvement

Flagging &
Communications
Marshal for SCCA and
the Circuit of the
Americas, 08/25 – Present

LANGUAGES

English – Fluent
Sinhala – Fluent
Japanese – Basic
Spanish – Basic